

3 Directory structure and filesystem

3.1 Directory structure

Let us get familiar with the directory structure of your Ubuntu set-up. The file system is typical Unix, with a hierarchal structure, best understood as a tree-like structure for those of you familiar with algorithmic design. The top of the tree, i.e., the highest level in the file system is the root directory, or / and everything, including hard-disks, partitions, USB drives, directories and data files fall below the / as individual files. For windows users, the important thing to understand here is that unlike Windows, even hard disks and removable media devices are treated as files, not some special category, so you get more power in configuring all your devices.

The root directory of Ubuntu contains some important system directories, which are common with most other Linux distributions:

- **bin** – This directory contains important binary applications
- **boot** – This directory contains the files needed for booting up the operating system
- **dev** – This directory contains the hardware device files that we will talk in detail in a later section
- **etc** – This is the place where you can find the configuration files of the base OS and other installed applications and startup scripts
- **home** – This is your home directory where all your personal data is stored
- **lib** – The system libraries required for the proper functioning of installed software are present in this directory
- **lost+found** – As the name suggests, this contains the lost and found files of your / directory.
- **media** – This directory is where all your mounted (loaded) removable media such as CDs and digital cameras will figure up as files, for you to configure
- **mnt** – This directory contains the mounted filesystems of your computer
- **opt** – This directory provides a location for optional applications to be installed
- **proc** – It's a special dynamic directory that maintains information about the state of the system, including currently running processes and their thread specific details
- **root** – This is commonly known as the slash-root directory, because we refer to / as root in Ubuntu. It's nothing, but the root user's home directory
- **sbin** – Some important system binaries are stored here
- **srv** – This directory acts as a temporary location for data meant to be used by servers
- **sys** – This directory contains system-specific information meant as